

FT LEONARD WOOD, MO, USA

WMO: 724457

Lat: 37.740N

Lon: 92.140W

Elev: 353

StdP: 97.15

Time Zone: -6.00 (NAC)

Period: 07-19

WBAN: 03938

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.5	-11.3	-20.1	0.7	-13.0	-17.8	0.8	-9.3	9.2	3.5	8.4	4.0	2.7	320	0.452

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	34.9	24.1	33.2	24.0	31.9	24.0	26.2	31.8	25.5	30.9	24.8	30.0	2.7	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.7	20.6	29.0	24.0	19.7	28.4	23.3	18.9	27.7	83.2	31.8	80.2	31.1	77.1	30.0	28.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.4	7.5	6.8	-18.4	36.4	2.8	1.7	-20.4	37.7	-22.0	38.7	-23.6	39.6	-25.6	40.9
			-18.8	27.2	2.7	0.9	-20.8	27.8	-22.4	28.3	-23.9	28.8	-25.9	29.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	13.6	0.4	2.6	8.5	14.0	18.5	23.7	25.5	24.0	20.6	14.0	7.9	2.6		
	DBStd	10.04	6.83	6.85	6.27	5.12	4.73	3.07	3.01	3.07	3.98	5.10	5.46	5.60		
	HDD10.0	1008	303	222	103	20	2	0	0	0	0	20	101	236		
	HDD18.3	2489	555	441	309	147	59	2	0	1	22	151	314	488		
	CDD10.0	2316	7	16	56	139	267	411	480	435	317	145	38	7		
	CDD18.3	756	0	1	4	16	65	163	222	178	89	18	1	0		
	CDH23.3	7169	0	3	40	160	563	1542	2289	1644	774	146	7	0		
	CDH26.7	2838	0	0	4	31	151	607	1059	687	274	25	0	0		
(14) Wind	WSAvg	3.1	3.4	3.5	3.7	3.8	3.1	2.7	2.3	2.3	2.5	3.0	3.4	3.3		
(15) Precipitation	PrecAvg	1084	57	57	91	109	131	110	103	98	108	78	85	66		
	PrecMax	1559	148	152	254	303	290	195	263	357	344	275	231	273		
	PrecMin	648	13	4	30	16	25	10	19	20	9	6	9	16		
	PrecStd	249	32	40	49	56	60	47	59	61	72	50	60	53		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.8	23.1	27.1	30.1	32.1	35.1	38.0	36.3	33.7	29.4	24.5	19.5		
		MCWB	13.8	15.5	17.0	19.1	21.1	23.7	23.1	24.1	23.2	20.6	16.4	15.7		
	2%	DB	16.9	19.4	24.0	27.2	29.9	32.9	35.7	34.1	31.9	26.9	21.5	16.8		
		MCWB	11.9	13.5	15.7	17.6	21.4	24.7	23.8	24.0	23.4	18.5	14.3	13.6		
	5%	DB	13.6	16.7	21.1	25.1	28.4	31.7	33.6	32.6	30.1	25.0	19.1	14.0		
		MCWB	9.1	11.8	14.0	16.5	20.9	24.2	24.5	23.5	22.8	17.8	13.7	9.9		
	10%	DB	10.6	13.5	18.6	22.9	26.7	30.3	32.0	30.6	27.9	22.6	16.9	11.5		
		MCWB	6.7	9.4	12.5	15.9	20.0	23.6	24.5	23.3	21.5	16.7	12.1	8.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.2	16.8	18.8	20.9	23.9	26.8	27.1	27.0	25.2	22.4	18.3	16.9		
		MCDB	18.6	21.5	24.3	26.5	28.9	31.6	31.9	32.9	30.5	26.9	21.6	18.9		
	2%	WB	13.0	15.2	17.1	19.6	22.9	25.7	26.2	25.9	24.2	20.8	16.4	14.4		
		MCDB	14.9	18.8	22.0	23.9	27.9	30.7	31.8	31.8	29.6	24.2	19.2	16.1		
	5%	WB	9.9	13.1	15.4	18.5	21.9	25.0	25.6	25.0	23.5	19.3	14.8	11.3		
		MCDB	12.5	15.3	19.9	22.8	26.2	29.9	31.1	30.5	28.5	22.6	17.9	13.1		
	10%	WB	6.9	9.6	13.8	17.1	21.2	24.2	24.9	24.1	22.6	18.0	12.8	8.8		
		MCDB	10.6	13.1	17.3	21.2	25.3	28.9	30.2	28.9	27.0	21.6	16.2	10.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.6	10.6	11.5	12.0	11.3	10.7	11.0	10.9	11.4	12.0	11.2	9.6		
		MCDBR	14.1	14.6	14.9	14.4	12.8	12.0	13.1	13.3	12.7	13.3	13.9	13.5		
	5% WB	MCWBR	9.5	9.8	7.7	7.2	5.7	5.3	4.2	4.7	5.2	6.6	8.4	9.3		
		MCWBR	12.7	12.9	12.9	12.1	10.8	10.8	10.9	11.1	11.1	11.1	11.8	11.9		
(40) Clear-Sky Solar Irradiance	taub		0.300	0.308	0.336	0.384	0.415	0.426	0.452	0.450	0.386	0.344	0.309	0.295		
	taud		2.482	2.457	2.426	2.298	2.272	2.264	2.223	2.222	2.378	2.486	2.532	2.517		
	Ebn at Noon		889	927	928	895	869	856	830	823	863	873	873	872		
	Edh at Noon		80	93	106	128	134	135	140	137	111	90	76	73		
(44) All-Sky Solar Radiation	RadAvg		2.27	2.90	3.88	5.03	5.65	6.45	6.27	5.61	4.84	3.52	2.45	1.92		
	RadStd		0.23	0.34	0.33	0.31	0.49	0.49	0.43	0.32	0.33	0.38	0.26	0.22		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (21 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page